

Maury High School Network Infrastructure Project Proposal

Background: Maury High School in Norfolk, Virginia, needs a proper network setup.

Objective: Design a Network Infrastructure Consisting of:

- Plan and budget for wiring (outlets, cable runs)
 - Floor plan (locations of closets and equipment rooms)
 - Plan and budget for Ethernet network (cables and switches)
 - Plan and budget for securing network traffic (firewall, subnets/VLANs)
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1. Building Overview:

- Three floors plus one basement
- Assumed to be 300 feet wide. Scale: 828 pixels per 300 feet (0.362 ft/px)
- Assume 9-foot floors (13 floor-to-floor)
- Four small offices on each floor pre-approved for use as wiring spaces
- Up to four telecommunications closets (TCs) may be used (one per floor)

2. Equipment Room (ER) and Telecommunication Closets (TCs):

- Main Equipment Room (ER): Basement, lower-left office. This is the MDF
- Telecom Closet 1 (TC1): First floor small office (lower left). IDF1
- Telecom Closet 2 (TC2): Second floor small office (lower left). IDF2
- Telecom Closet 3 (TC3): Third floor small office (lower left). IDF3

3. Cabling Plan and Budget

3.1 Cabling Type

- Category 6 Ethernet (Cat6) cable (supports 1 Gbps)
- Patch panels and RJ45 jacks rated for Cat6

3.2 Calculated Cable Runs

- Scale-based distance: 300 feet = 828 pixels
- Cable runs from TC to offices (per floor): $1712 \text{ px} = 619 \text{ ft} * 2(\text{outlets}) = 1238 \text{ ft}$. Add Slack: $1238 \text{ ft} * 1.10 = 1362 * 3 (\text{floors}) = 4086 \text{ ft}$
- 1st floor cable runs from TC to other rooms: $4945 \text{ px} = 1791 \text{ ft} * 2(\text{outlets}) = 3582 \text{ ft}$. Add slack: $3582 \text{ ft} * 1.10 = 3940 \text{ ft}$.
- 2nd floor cable runs from TC to other rooms: $5303 \text{ px} = 1920 \text{ ft} * 2(\text{outlets}) = 3840 \text{ ft}$. Add slack: $3840 * 1.10 = 4224 \text{ ft}$
- 3rd floor cable runs from TC to other rooms: $4674 \text{ px} = 1692 \text{ ft} * 2(\text{outlets}) = 3384 \text{ ft}$. Add slack: $3384 * 1.10 = 3722 \text{ ft}$.
- Basement runs from ER: $747 \text{ px} = 271 \text{ ft} * 2(\text{outlets}) = 542 \text{ ft}$. Add slack: $542 * 1.10 = 596 \text{ ft}$
- Vertical riser cables between ER and each TC: 13 ft each = 39 ft total. Add slack: $39 * 1.10 = 43 \text{ ft}$.

Plan:

Destination	Cable run (ft)	Rooms	Total (ft)
Office runs	619	9	4086
First floor	1791	16	3940
Second floor	1920	16	4224
Third floor	1692	15	3722
Basement	271	3	596
Vertical cabling	13	3 rises	43

Total Cable Estimate: 16,620 ft.

3.2 Outlets and Wall Plates

- Standard layout: 2 Ethernet outlets per classroom/office
- Total number of rooms: 50

3.4 Hardware Cost

Item	Unit Cost	Quantity	Total Cost	Source
Cat6 bulk cable (1000 ft)	\$294.99	17	\$5014.83	Monoprice

Cat6 RJ45 keystone jacks (50)	\$29.99	2	\$59.98	Amazon - Rapink
Dual-port wall plates (20)	\$17.99	3	\$53.97	Amazon - FGB
24-port Cat6 patch panel	\$27.99	5	\$139.95	Monoprice

Total Hardware Cost: \$5270 (rounded up)

4. Ethernet Equipment Plan and Budget

4.1 Required Switch Ports

-Minimum ports: Outlets * 1.1 (must be at least 10% more than outlets)

4.2 Hardware

-48-port – 2 (uplinks) = 46 usable ports

-Managed 48-port gigabit switches: 50 rooms (two ports) = $100 * 1.10 = 110$ ports. $110 / 46 = 3$ switches.

-Rack-mounted UPS: equipment room (1) + TCs (3) = 4 UPS

-Short patch cables (1 ft each): $110 + 3$ (uplinks) = 113 ft

-Room patch cables (7ft each): $50 * 2 = 100$ 7ft patch cables

4.3 Equipment List

Equipment	Unit Cost	Quantity	Total Cost	Source
48-port managed switch	\$335.78	3	\$1007.34	Amazon - MokerLink
Rack-mounted UPS (APC 1000VA)	\$421.13	4	\$1684.52	Walmart - APC
1 ft patch cables (TC panels)	\$1.54	113	\$174.02	Monoprice
7 ft patch cables (rooms)	\$3.39	100	\$339.00	Monoprice

Total Equipment Cost: \$3205 (rounded up)

5. Network Security Plan

5.1 Firewall Requirements (200-300 users)

- VLAN support
- Policy-based blocking and filtering
- Gigabit throughput

5.2 Proposed Firewall

- Fortinet FortiGate 60F

5.3 Network Design (VLAN Plan)

VLAN	Purpose	Ex Subnet
VLAN 10	Staff	10.10.10.0
VLAN 20	Students	10.20.20.0
VLAN 30	Admin	10.30.30.0
VLAN 40	Servers	10.40.40.0

This will allow the following access:

- Staff: Internet
- Admin: All internal networks
- Students: Filtered internet access only

And block:

- Students: Staff and admin VLAN

5.4 Firewall Cost

Item	Cost	Source
Firewall appliance	\$641.00	Online Components

Total Cost: \$641.00

Total Cost of Project:

\$9116.00

Conclusion:

In this proposal, I provided a complete cost-effective network design for Maury High School. By implementing a structured cabling system and dedicated and proper TCs, the school will gain well-structured internet access. Additionally, paired with Ethernet capabilities and VLAN network architecture, Maury will be given a reliable foundation for both administrative and instructional technology. The chosen hardware ensures operability within reasonable cost ranges. With the proper installation, it is my hope that this design will be sufficient for all of Maury High School's internet needs.

Sources Extended (links)

Cat6 (1000ft): https://www.monoprice.com/product?p_id=8103

Cat6 RJ45 jacks: https://www.amazon.com/Rapink-Keystone-Support-Plate-Cabling/dp/B0CJ6JYQP7/ref=sr_1_1_sspa?crid=17RT07DBM18Z1&dib=eyJ2ljojMSJ9.eor1lMpS91PLuZ5BRUchKeVjQZh_fSsx98XqoiYRG8Hgl1qtoviP0iLPY-fS-PxZWQKMyeW-YoxrS28QNHnZDxmMrxr4VFxtnSRzCBvBU0OjZW3odQj-rbGXwo3m2H78TqbQUbrglfdb4BR4GjJJ4h1c3jTEnLF3U_Zz4_b8-VxxWDyzqT4gjwkXEE_U_vLRYGvPB1WHejr-6Tga1aOGSQPMJ1xMas6j6uV4_-3DYV4.gPD12dxN_yK9ICH6BadPizC-SyZgIM8bljB5A92uVpw&dib_tag=se&keywords=cat6%2Brj45%2Bkeystone%2Bjack%2Bwith%2Bwall%2Bplate&qid=1765331908&sprefix=%2Caps%2C61&sr=8-1-spons&sp_csd=d2lkZ2V0TmFtZT1zcF9hdGY&th=1

Dual-port wall plates: https://www.amazon.com/FGB-Keystone-20-Pack-Low-Profile-Inserts/dp/B0D9FHxDJ5/ref=sxin_17_pa_sp_search_thematic_sspa?content-id=amzn1.sym.f0e9aec8-2844-49ab-bbaa-8468b42715cd%3Aamzn1.sym.f0e9aec8-2844-49ab-bbaa-8468b42715cd&crid=2D5L2M955XQ4G&cv_ct_cx=cat6%2Bdual%2Bport%2Bwall%2Bplate&keywords=cat6%2Bdual%2Bport%2Bwall%2Bplate&pd_rd_i=B0D9FHxDJ5&pd_rd_r=e4d15f53-e5d7-44d6-a312-72faed677f62&pd_rd_w=eyZSu&pd_rd_wg=cj5uU&pf_rd_p=f0e9aec8-2844-49ab-bbaa-8468b42715cd&pf_rd_r=R2BJWYJXDR2K2ZVN665Z&qid=1765331962&sbo=RZvfv%2F%2FHxDF%2BO5021pAnSA%3D%3D&sprefix=%2Caps%2C169&sr=1-1-6024b2a3-78e4-4fed-8fed-e1613be3bcce-spons&aref=fYxLEXVKC9&sp_csd=d2lkZ2V0TmFtZT1zcF9zZWZyY2hfdGhlfWF0aWM&th=1

Patch panel: https://www.monoprice.com/product?p_id=7253

Managed switch: https://www.monoprice.com/product?p_id=7253

UPS: https://www.walmart.com/ip/APC-1000VA-Smart-UPS-with-SmartConnect/256112359?wmlspartner=wlpa&selectedSellerId=0&utm_source=chatgpt.com

1 ft cables: https://www.monoprice.com/product?p_id=9819

7 ft cables: https://www.monoprice.com/product?p_id=9819

Firewall: https://www.onlinecomponents.com/en/productdetail/fortinet/fg60f-56640994.html?utm_campaign=searchads360feed&utm_content=instock&utm_term=FG-60F&gclid=aw.ds&gad_source=1&gad_campaignid=22426139663&gbraid=0AAAAADkP1yYDs2B0PGnW5n0X0ujW-9Jgu&gclid=Cj0KCQiArt_JBhCTARIsADQZayn4DMjQW_JNF9MDXvxE8tWXJuzcV2WF817dHMqe-aS97Os3Q-mWnqkaAlzqEALw_wcB